

Prey-predator system

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Abstract

Predator-prey interactions shape ecological dynamics, and the Lotka-Volterra model provides a foundational framework for this study. This report explores its application to the Canadian lynx and snowshoe hare system, critically evaluating key assumptions that influence model accuracy. By comparing various modified models against historical data, we assess their validity and effectiveness in representing real-world dynamics. Furthermore, we integrate multiple modified systems to develop a more accurate representation of predator-prey interactions. This study provides insights into improving ecological modelling and to a deeper understanding of population dynamics in natural ecosystems.

1 Introduction

Predator-prey relationships are fundamental to ecological systems and influence population dynamics and ecosystem stability [1]. In this report, we examine the Canadian lynx and the Snowshoe hare, a predator-prey pair found in the boreal forests of northern Canada. The lynx depends heavily on hares as its primary food source, with hares making up 96% of its diet [2]. Hare populations are affected by food availability, predator pressure, and environmental conditions [3] [4]. The hare population faces seasonal challenges, particularly during winter, when vegetation is scarce and they rely on their white fur for camouflage [5]. As the numbers of hares decrease, lynx populations also suffer due to food scarcity, leading to a lagging decline in predator numbers [6]. However, lynx can supplement their diet with small mammals and birds, which helps to support them during hare population crashes [2].

The Lotka-Volterra model serves as our foundational model, which provides fundamental insights into predator-prey interactions [7]. However, real-world ecosystems are far more complex, with external factors such as food scarcity, alternative prey, and environmental variability that influence population dynamics. Understanding these interactions is crucial for ecological conservation and wildlife management, as it

helps predict population fluctuations and informs sustainable conservation strategies. Our report aims to examine the Lotka-Volterra model by developing modified models that address its key assumptions, which may limit its applicability to real-world ecosystems. Through comparative analysis of models and historical data [8], we quantitatively and qualitatively evaluate the strengths and limitations of our own modified models in representing the complex population dynamics of lynx and hare. By refining the Lotka-Volterra model, we aim to deepen our understanding of predator-prey systems and contribute to more accurate ecological modelling.

The assumptions we investigated were as follows:

1. Non-exponential growth: In the absence of predators, the prey population would grow proportionally to its size [7].
2. No age distribution: The reproductive rate of the population is independent of the age distribution of the population [4].
3. Unlimited food: The prey have an unlimited food supply at all times [1].
4. No alternative to prey: In the absence of prey, the predator population declines exponentially, without alternative food sources to sustain it [1].
5. Inevitable Predator-Prey interaction: When both predators and prey are present, there will be a decrease in the prey population and a corresponding increase in the predator population. [7].

2 The Lotka-Volterra model

2.1 Lotka-Volterra model

The Lotka-Volterra model consists of two ordinary differential equations that describe the population growth rates of hares and lynxes. It can be understood as follows: the rate of change in the hare population is determined by its natural growth rate minus the rate at which lynxes prey on them. Likewise, the change in the lynx population is driven by the availability of hares as a food source, offset by the lynxes' natural mortality rate.

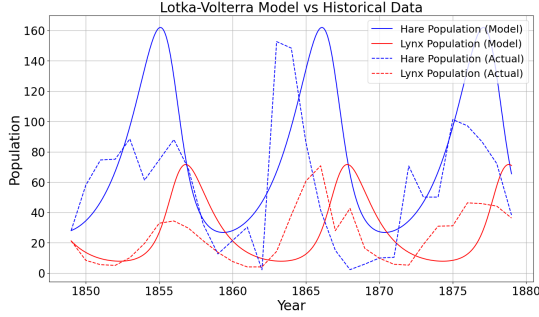


Figure 1: The Lotka-Volterra model captures the cyclic predator-prey dynamic, like lynx and hare populations, with its characteristic oscillating shape that is also seen in the empirical data. As lynx thrive, hare numbers drop due to predation, leading to a lynx decline and hare recovery, aligning with historical data.

$$\frac{dH}{dT} = \alpha H - \beta HL, \quad (1)$$

$$\frac{dL}{dT} = -\gamma L + \delta HL. \quad (2)$$

2.1.1 Coefficients:

α = The natural growth rate of the prey or the auto-increase. It is the growth in population per capita without predation.

β = The rate at which predators consume prey.

δ = Efficiency of converting consumed prey into predator growth.

γ = Natural death rate of predators

The equations show the rate of population growth for the hare and for the lynx.

The data used in Figure 1 and throughout this report is extracted from a graph in Odum's *Fundamentals of Ecology* (p. 191), sourced in MacLulich's 1937 study *Fluctuations in the Numbers of Varying hare* [8, 9]. The data represents the number of pelts traded at Hudson's Bay, rather than actual animal population counts. A significant portion of these pelts were contributed by Native Americans, and historical medical outbreaks during this period may have influenced the data, introducing potential biases. However, since this was the only large dataset available to us, it is used throughout the report.

2.1.2 Assumptions of Lotka-Volterra model:

This report only investigates some assumptions of the basic Lotka-Volterra model. Here is a list of the assumptions that this report **does not** explore:

1. **Homogeneous Environment:** The environment remains uniform, which means that external factors such as seasonal changes, habitat variation, disease, or human interference do not affect predator-prey interactions [8] [5].

2. **Closed System:** The model assumes no immigration or emigration of individuals, which means that external influences such as new predators, prey migration, or human intervention are not considered [8].
3. **Constant Interaction Coefficients:** The rates of predation and prey reproduction are fixed, despite real-world variations due to environmental and behavioural changes [3].
4. **No Intraspecific Competition:** Prey and predator populations do not experience competition within their own species, in reality, individuals compete for food, shelter, and mates [1].
5. **Constant Predator Efficiency:** Predators capture and consume prey at a fixed rate, but real-world predators adapt their hunting strategies and may experience decreased efficiency at low prey densities [6].

2.2 Assumption 1 - Exponential growth of the prey population in the absence of predators:

2.2.1 Fitting the Assumption to the Model

The ordinary differential equation for the hare population growth rate in the Lotka-Volterra model is:

$$\frac{dH}{dT} = \alpha H - \beta HL. \quad (3)$$

Equation 3 consists of two terms. The first term, represents Malthusian growth — the per capita growth rate of the hare population under ideal conditions, multiplied by the current population size [article]. The second term accounts for predation. Removing the predation term gives the Malthusian growth model:

$$\frac{dH}{dT} = \alpha H \quad (4)$$

Integrating this ODE with respect to time gives:

$$H(t) = H_0 e^{(\alpha t)} \quad (5)$$

Equation 5 shows that the hare population grows exponentially in the absence of predators. However, this is unrealistic for large values of time, as unlimited growth is not feasible. Sustaining a population requires sufficient resources such as food, shelter, and water [7].

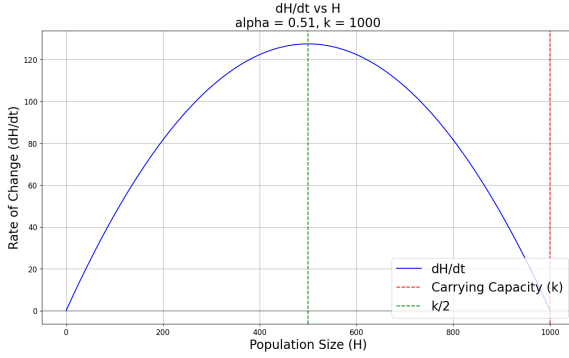
2.2.2 Creating a Modified Model

To fix the inaccurate assumption of the Lotka-Volterra model we need to mathematically incorporate a term that relates the finite resources of the ecosystem to the population growth. Incorporating a density-dependent population control into the growth equation

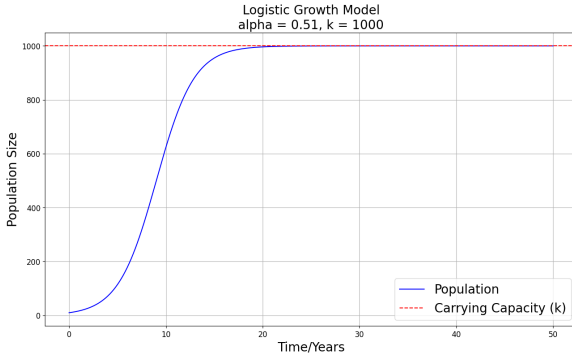
we can model the maximum limit of the hare population size. Adding the damping term $(1 - (H/K))$ gives the Logistic growth model for hare population in the absence of predators:

$$\frac{dH}{dt} = \alpha H \left(1 - \frac{H}{K}\right) \quad (6)$$

This model assumes that there is a maximum population size that the environment can support, expressed as, K , the 'Carrying-capacity'. [10] This parameters value may fluctuate with the chaotic variability of the environmental conditions but for our model we assume



(a) Population growth rate with resource limitations. Growth peaks when .



(b) Population vs. Time for logistic growth. Shows equilibrium being reached in the absence of predation.

Figure 2: Logistic growth model.

Mathematically, at equilibrium:

1. $\frac{dH}{dt} = \alpha H - \alpha H^2/K = 0$
2. $\alpha H = \alpha H^2/K$
3. $K = H$

Incorporating the predation term from the Lotka-Volterra model yields the modified logistic growth equation [10] gives:

$$\frac{dH}{dt} = \alpha H \left(1 - \frac{H}{K}\right) - \beta HL \quad (7)$$

2.2.3 Results of Logistic growth model

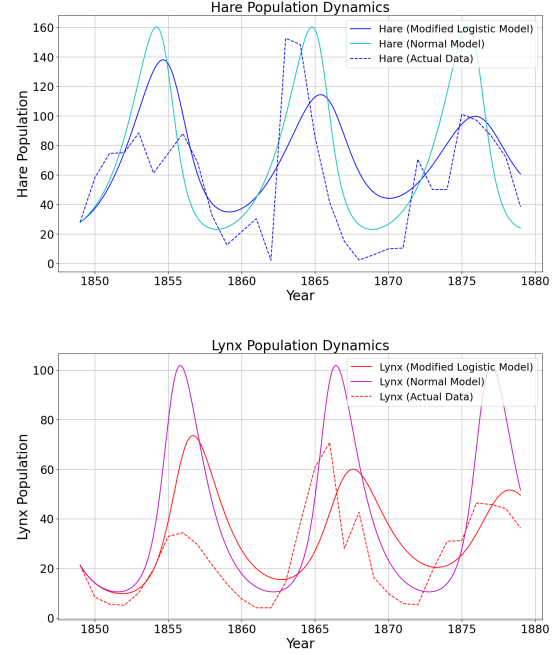


Figure 3: Comparison of the modified Model, the Basic model, and real-world data for the Hare and Lynx population. This plot illustrates how each model aligns with Hare and Lynx population trends.

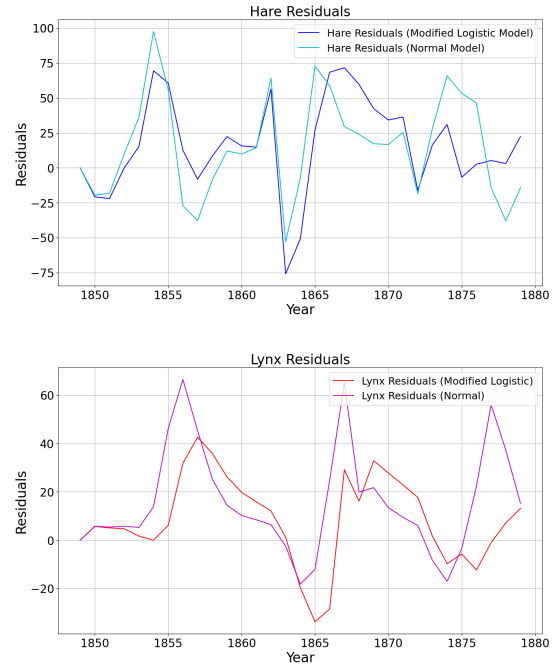


Figure 4: Residuals and absolute residuals for the Hare and Lynx population. RMS values for Hare population: Modified Model = 37.18, Lotka-Volterra Model = 39.40. RMS values for Lynx population: Modified Model = 19.98, Lotka-Volterra Model = 26.7

In both cases, modifying the Lotka-Volterra model improves the accuracy, as seen by the improved RMS values of the residuals. This effect is further pronounced for lower predation rate, suggesting that such modifications become even more valuable when applied to different species or unusual ecological conditions. This highlights a key limitation of the classical Lotka-Volterra model. However, when comparing the plotted results of both models against the empirical data, it is evident that the original Lotka-Volterra model, under normal conditions, effectively captures the characteristic cyclic dynamics of predator-prey systems, such as those observed in the hare and lynx populations.

2.2.4 Assumptions of the Modified Model

The modified model replaces the assumption of exponential prey growth with logistic growth while retaining other Lotka-Volterra assumptions. Additional assumptions include:

1. The carrying capacity remains constant over the studied period.
2. The predator-prey interaction remains unchanged.
3. No external environmental disturbances significantly impact population dynamics.

2.3 Assumption 2 - Age responsive Lotka-Volterra Model:

Age distribution is not an external factor itself; however, it serves as a bridge between external factors (weather, habitat, etc.) and population dynamics of both predator and prey.

$$H = H_a + H_j \quad (8)$$

where H_a is the adult hare population and H_j is the juvenile hare population, L_a is the adult lynx population, and L_j is the juvenile lynx population

$$H_a = {}_H H, H_j = (1_H)H \quad (9)$$

$$L_a = {}_L L, L_j = (1_L)L \quad (10)$$

where η_H and η_L are the proportion of adults in the lynx and hare populations at a given time.

This is helpful, since only adult hares reproduce; therefore, the reproduction term αH can be written as $\alpha \eta_H H$. Similarly, only adult lynx contribute to the growth of the population through hunting therefore, the growth term for lynx can be written as $\delta \eta_L H L$.

Canadian lynx appear to have more preference for the Adult population of Snowshoe hares. Juveniles are predated at a higher rate or vice versa. we can consider r to be the preference factor for juveniles. Therefore, the total predation term $\beta H L$ can be written as $\beta H L (\eta_H + r(1 - \eta_H))$ and where $r = \frac{\text{JuvenilesPreferred}}{\text{AdultsPreferred}}$

Using The Klaun Book[11] The data given indicates that only 38 percent of predation on juvenile hares is attributable to the lynx. In contrast, 82 percent of adult hare mortality is due to predation, with 65–75 percent of these deaths caused by mammalian predators, primarily lynx. These findings suggest that lynx tend to preferentially prey on adult hares.

It is important to note that this preference may be influenced by external factors such as disease, habitat changes, or changes in prey availability, and could vary over time or across regions. Which is why we think, the inclusion of age as a factor in this ODE is important

For lynx, the mortality term, γ can vary among juveniles and adults, so we can separate it to γ_a and γ_j for adults and juveniles respectively. so it can be written as: $(\gamma_j(1 - \eta_L) + \gamma_a \eta_L) L$

According to the forest department of the United States Forest Services report [12] Dividing the mortality term γ by age classes could provide more accurate information on the impact of external factors on the mortality of the Canada lynx. So we can modify our Lotka-Volterra equations:

$$\frac{dH}{dt} = \alpha \eta_H H - \beta H L (\eta_H + r(1 - \eta_H)), \quad (11)$$

$$\frac{dL}{dt} = -(\gamma_j(1 - \eta_L) + \gamma_a \eta_L) L + \delta \eta_L H L. \quad (12)$$

2.3.1 Assumptions Added

1. Age distributions remain stable over time; therefore, we assume constant η_H and η_L .
2. Populations are isolated in a given space and do not migrate into or out of this space.

2.3.2 Analysis of age-structured model

Parameters:

1. Preference ratio ($r = 0.543$): From the field study in boreal forests, lynx are responsible for 38% of juvenile hare predation and 70% of adult hare predation, giving $r = \frac{0.38}{0.70} = 0.543$, reflecting a preference for adults [11].
2. Mortality rates ($\gamma_j = 0.460$, $\gamma_a = 0.793$): Juvenile lynx mortality (46. 0%) is derived from a survival rate 54% (1990–2000 data); adult mortality (79.3%) due to trapping impacts by humans, and since the data used are derived from similar factors, it can be helpful to better model [12].
3. ($\eta_H = 0.225$, $\eta_L = 0.587$): Juvenile hares average 77.5% ($\eta_H = 1 - 0.775$), and juvenile lynx comprise 42.3% ($\eta_L = 1 - 0.423$). Stable age distributions justify constant η_H and η_L [11, 12].

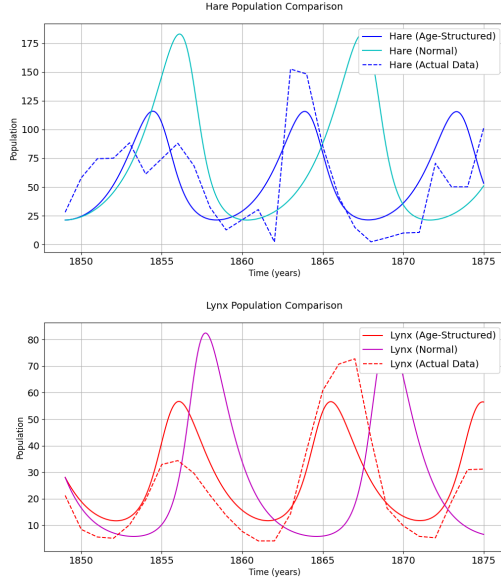


Figure 5: Comparison of the Age-Structured model, the Basic model, and observed real-world data for the Hare and Lynx population. The Age-Structured model incorporates age-specific dynamics, while the Basic model uses a simpler approach.

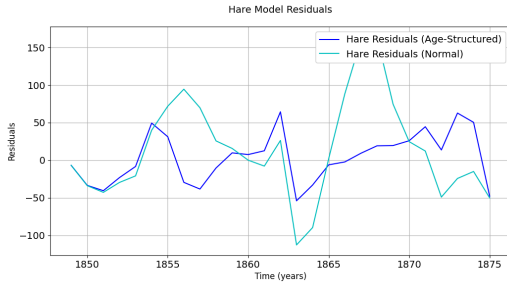


Figure 6: Residual analysis for the Age-Structured and Basic models. The root mean squared residual for the Age-Structured model is 33.02, compared to 39.40 for the Basic model. Although the difference is modest, the Age-Structured model's residuals fluctuate closer to zero, suggesting a slightly better fit to the observed data, as evidenced by the residual plot.

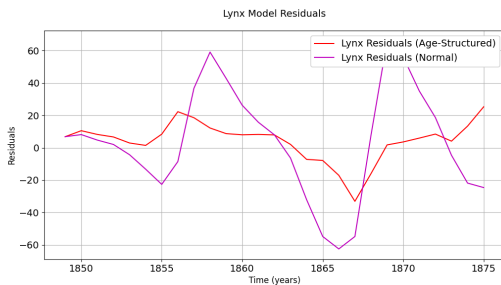


Figure 7: The Age-Structured model is visibly much better than the regular model, with root mean squared residuals of 12.90 compared to 26.73 for the basic model, making the Age-Structured model superior. The peak and troughs of the age structured model also match the real life data better than the normal model.

The age structured model improves the accuracy of the model by incorporating age-dependent predation preference, mortality and reproduction, aligning with empirical data. The age-structured model appears to be damped and has lower peaks, as the reproductive factor is only limited to adult populations. Because $r < 1$, it suggests that adult hares are predated on more, which means that, after a population decline, it takes a delay to rebound to the peak level, ending up with a better match to historical data.

2.4 Assumption 3 - The prey having infinite food at all times

The best way to adapt the base Lotka-Volterra model for the prey not having infinite food is to treat this as an extension of the predator-prey ODE's. This means having another ODE linking the prey to the food it eats, the same way we have a differential equation between the predator and prey. The Snowshoe hare has a diet that varies by season however they mainly eat vegetation[nps'Snowshoe'hare]. This means that we can define a variable V for the vegetation and create an ODE

$$\frac{dV}{dt} = rV\left(1 - \frac{V}{G}\right) - \lambda HV. \quad (13)$$

2.4.1 Coefficients:

λ = Rate that prey consume vegetation

r = Growth rate of vegetation

G = Maximum carrying capacity of vegetation (the maximum amount of vegetation that the environment can sustain).

The differential equation dV/dt is affected by two key processes:

1. Natural regeneration of vegetation. This is shown by $rV\left(1 - \frac{V}{G}\right)$.
2. Consumption of vegetation by the prey shown as $-\lambda HV$.

2.4.2 Final ODE's:

As we have now introduced variable V , we can adapt the initial base Lotka-Volterra equations so the prey equation is now:

$$\frac{dH}{dt} = \alpha HV - \beta HL, \quad (14)$$

and the predator equation stays unchanged. In order to compare to real life data, it is beneficial to combine these three ODE's into two. Assuming vegetation in the boreal forest stays relatively constant over time, $\frac{dV}{dt} = 0$. As $V \neq 0$ (since vegetation cannot be zero), solving for V :

$$V = G\left(1 - \frac{\alpha H}{r}\right). \quad (15)$$

Substitute V into the Prey and Predator Equations

Prey Equation:

$$\frac{dH}{dt} = \alpha GH - \frac{\alpha^2 GH^2}{r} - \beta HL \quad (16)$$

As predator equation does not have any V terms, it remains unchanged:

$$\frac{dL}{dt} = -\gamma L + \delta HL. \quad (17)$$

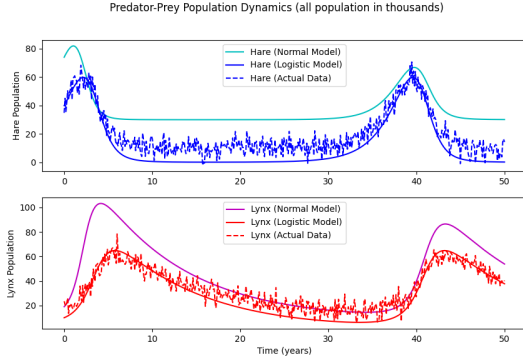


Figure 8: This graph presents the population dynamics of hares and lynx over time, comparing actual observed data (dashed lines) with two theoretical models: the Logistic Model and the Lotka-Volterra Model. The Logistic Model is more realistic than Lotka-Volterra and as a result has a line that fits with the actual data better. This clearly shows how tackling the assumption of prey having infinite food has improved the model.

2.4.3 Assumptions of the Modified Model:

- 1) We assume that $dV/dt = 0$ in order to create two ODE's instead of three ODE's. This assumption makes it easier to model against real life data. On the most part, the assumption is fair as for most the year rate of vegetation growth will be fairly constant, but during season changes and times of rapidly changing amounts of vegetation, this assumption is no longer valid.
- 2) We also assume that vegetation can never be zero. For most of the year this assumption is fair however there are extreme conditions where the vegetation would get close to zero for example a forest fire or a flood. Even during these periods, vegetation will never truly be zero but the vegetation accessible to the hares might be about zero, so this model could be further adapted to work under extreme conditions.

2.5 Assumption 4 - Lotka-Volterra Model with the predators having infinite appetite

In the basic form of Lotka-Volterra, it assumes that the predators will constantly be able to eat without getting

full. In reality, this wouldn't happen as the predators will inevitably become saturated if they have the ability to eat as much as they want. This would result in a lower predation rate during periods of time where prey population is high. We can adapt Lotka-Volterra by altering the predation term (βHL) and introducing either a linear, hyperbolic or a sigmoidal term - which depends on the behaviour of the predator and its' relationship with the prey. To work out this relationship we used the theory of Functional Response, which describes the relationship between a predator's feeding rate and the density and abundance of its' prey. This was developed into 3 types by C.S Holling [1], which showed these different types of relationships in linear, hyperbolic and sigmoidal form.

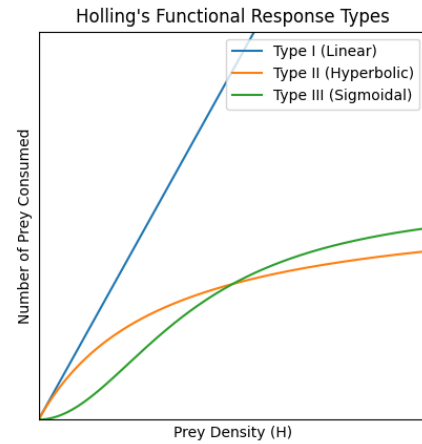


Figure 9: Holling's Functional Response Types

Type I (linear) is a very basic representation of functional response, which Lotka-Volterra is based on. The Canadian lynx and Snowshoe hare relationship has traits of both Type II (hyperbolic), as it follows saturation pattern due to handling time limits, and Type III (sigmoidal), as hares burrow and can camouflage in snow, making it harder and the lynxes burn more energy during the catch. However, due to many variables in handling time, prey capture effort, and the overall observed behavioural dynamics of the lynx and hare, the lynx follows more traits of Type II.

2.5.1 Applying to basic model

To apply this to the basic Lotka-Volterra Model, we need to first start with the Type II equation [13]:

$$f(H) = \frac{aH}{1 + ahH} \quad (18)$$

We then need to tailor this to the lynx and the hare, by finding the capture rate (a) and the handling time (h). By using $\alpha = 1.114$ as an estimate for Natural Growth Rate of hares, we can find the number of hares

that are consumed at each prey density:

$$f(H) = \frac{\alpha H_0 - (H_1 - H_0)}{L_0} \quad (19)$$

We can estimate a and h , by fitting the Holling's Type II equation to the computed values of $f(H)$ and H by using the nonlinear least squares approach to minimise the error:

$$\min_{a,h} \sum_i (f(H_i) - \frac{aH_i}{1+ahH_i})^2 \quad (20)$$

This optimisation finds the values of a and h that make the theoretical model match the observed data as closely as possible, which results in: $a = 0.1106$, $h = 0.1096$

Substitute back into Lotka-Volterra:

$$\frac{dH}{dt} = \alpha H - \left(\frac{aH}{1+ahH}\right)L = 1.114H - \left(\frac{0.1106H}{1+0.01212H}\right)L \quad (21)$$

$$\frac{dL}{dt} = -\gamma L + \delta \left(\frac{aH}{1+ahH}\right) = -\gamma L + \delta \left(\frac{0.1106H}{1+0.01212H}\right)L \quad (22)$$

2.5.2 Results - Functional Response

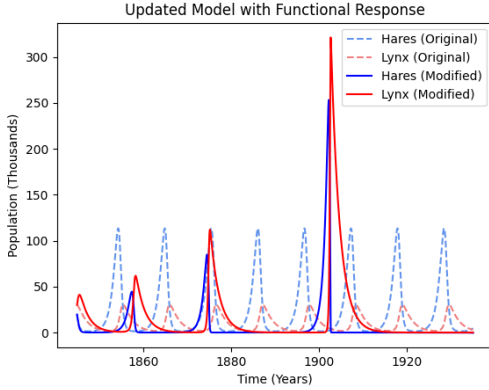


Figure 10: Updated Lotka-Volterra Model with Holling's Type II Functional Response

Figure 9 and 10 shows the effects of the hyperbolic term on the basic Lotka-Volterra, as it causes the unrealistic peaks and troughs in the cycle and causes it to increase exponentially [14]. This is due to the lowered predation rate for high prey densities, as the hare population now grows at steeper rate. This is followed by a rapid rise in the lynx population as now the individual lynxes eat less hares, so the system can support a larger lynx population as there is now enough hares to sustain it. This then also reaches an unrealistic peak, resulting in a sudden crash in hare population and then, in turn, the lynx population crashes.

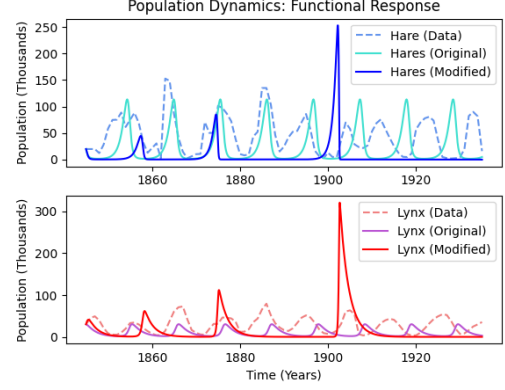


Figure 11: Predator Prey Population Dynamics for the Updated Model with Holling's Type II Functional Response

2.5.3 Improving with Non-linear Growth of Prey

To stop the model from constantly increasing, we can update the prey equation by adding the term: $(1 - \frac{H}{K})$ where the constant K is the carrying capacity of the hares, which is the maximum number of hares the environment can support. An estimate for K can be found within the data as the maximum number of hares recorded (152.65 thousand). Substituting into the model:

$$\frac{dH}{dt} = \alpha H \left(1 - \frac{H}{K}\right) - \left(\frac{aH}{1+ahH}\right)L \quad (23)$$

$$\frac{dL}{dt} = -\gamma L + \delta \left(\frac{aH}{1+ahH}\right)L \quad (24)$$

2.5.4 Results - Functional Response and Non-linear Growth

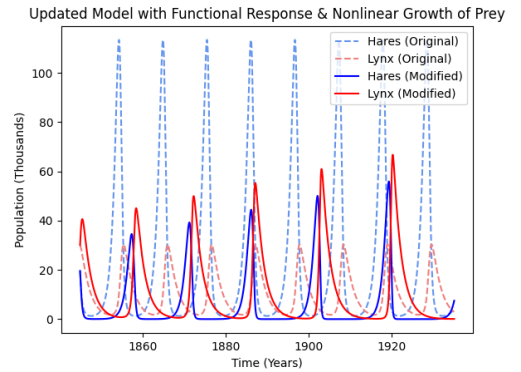


Figure 12: Updated Lotka-Volterra Model with Holling's Type II Functional Response and Non-linear Growth of Prey

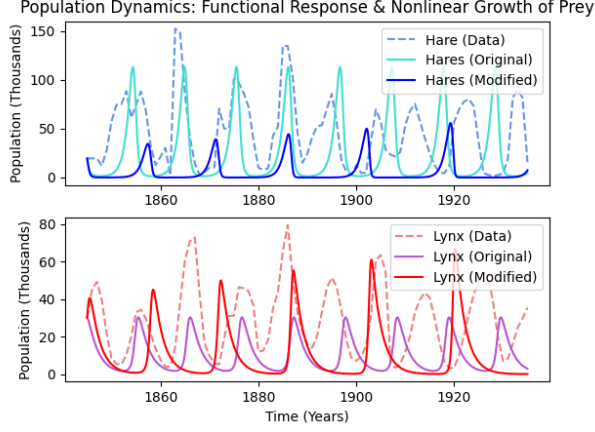


Figure 13: Predator-Prey Population Dynamics for the Updated Model with Holling's Type II Functional Response and Non-linear Growth of Prey

Figure 11 and 12 show how the added non-linear growth of prey reduces the extreme peaks of the population by allowing them to rise and fall in a more controlled manner. This is because the model introduces a natural population limit, which limits how fast hares can grow considering food availability and environmental constraints. This means that the hare population does not suddenly increase rapidly, which prevents the predator population from rapidly increasing in response and, ultimately, stops the model from crashing. Overall, this is a more stable model that is a closer representation of real-world predator-prey interactions.

2.5.5 Assumption of Modified Models

The two modified models add additional assumptions and build on the previous Lotka-Volterra assumptions. The additional and improved:

1. The hares grow exponentially at a constant rate α
2. The lynx consume hares at a saturation rate (Holling's Type)
3. The carrying capacity K is constant
4. The population growth rate immediately adjusts based on population size, there is no time delays

2.6 Assumption 5 - Spatial Lotka-Volterra Model with a finite range

The basic Lotka-Volterra assumes that the population of each species is distributed normally over their environment. This does not provide insight into the actual spatial distribution of the predator and prey species.

To implement spatial representation, the spatial domain was simplified as a 1-D strip with two edges defining the boundaries since it's simpler and more general

to analyse. Then, the population size H and L are converted to population densities h and l . This describes how population changes at each spatial point on the 1-D strip.

The diffusion was modelled using the following equation, also known as the heat equation:

$$\frac{\partial h}{\partial t} = D_h \left(\frac{\partial^2 h}{\partial x^2} \right) \quad (25)$$

where:

1. D_h is the diffusion coefficient
2. $\frac{\partial^2 h}{\partial x^2}$ measures how the population changes in space. It smooths out sharp variation by spreading individuals to other areas.
3. x is the coordinate in space.

2.6.1 Spatial Lotka-Volterra Equation

Implementing the diffusion equation into the Lotka-Volterra:

$$\frac{dh}{dt} = \alpha h - \beta h l + D_h(t) \frac{\partial^2 h}{\partial x^2}, \quad (26)$$

$$\frac{dl}{dt} = -\gamma l + \delta h l + D_l(t) \frac{\partial^2 l}{\partial x^2}. \quad (27)$$

2.6.2 Assumptions

These are the assumptions of the 1D diffusion equation model:

1. The function $D_H(t)$ and $D_L(t) = 0.1t$
2. All interaction occur locally, so predation at an exact spatial position is only caused by the population density at that position.
3. Time and space are modelled as discrete variables, enabling the application of a finite difference approximation.
4. According to Fick's diffusion, the population will move from high density to low density region randomly. Laplacian estimation was implemented to calculate the 2nd order spatial derivative of population density.

$$\frac{\partial^2 l}{\partial x^2} \approx \frac{l_{i+1} - 2l_i + l_{i-1}}{\Delta x^2}$$

5. [15] The reflecting (Neumann) boundary conditions were introduced as following:

$$\begin{aligned} \left. \frac{\partial n}{\partial x} \right|_{x=0} &= 0, & \left. \frac{\partial n}{\partial x} \right|_{x=L} &= 0, \\ \left. \frac{\partial p}{\partial x} \right|_{x=0} &= 0, & \left. \frac{\partial p}{\partial x} \right|_{x=L} &= 0 \end{aligned}$$

which describes a habitat with reflecting barriers. The rate of population change at the boundary is zero, resulting in no individuals leave or enter the system at the edges.

In the model, prey are uniformly distributed at the beginning, with a slightly higher concentration around the centre ($x = 50$) due to Gaussian distribution. Predator will be introduced at the centre and diffuse in all direction to uniform concentration.

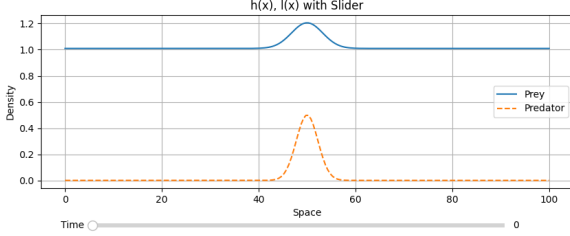


Figure 14: Initial distribution before prey and predator mix.

2.6.3 Results

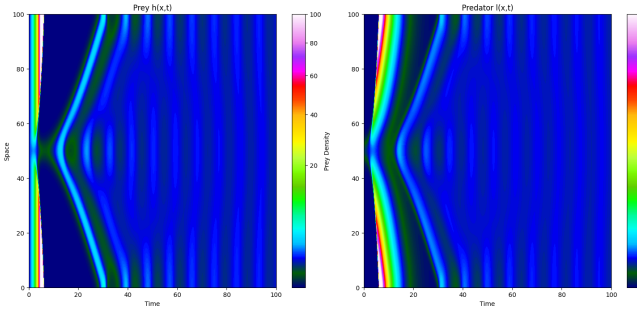


Figure 15: Population density contour of predator and prey along space time.

In Figure 15, the brighter bands indicate that there is higher concentration of prey or predator. Initially the prey's graph has higher density across the spatial points, it then spreads outward shown by a symmetric outward wave pattern. As time increases, the graph becomes darker and with more bands across the entire space. This represents the oscillation of the prey and predator's density of which the patterns become more diffused. From the predator graph, it implies that the predator does indeed chase the preys as they have very similar wave pattern but is lagging behind due to the time lag. The predator and prey graph are complementary to each other, where the darkest bands for prey correspond to the brightest bands for predator. As time goes on, the predator's density becomes more diffused and has reached a stable localised distribution. Besides that both predator and prey's population decrease and have the same behaviour over time.

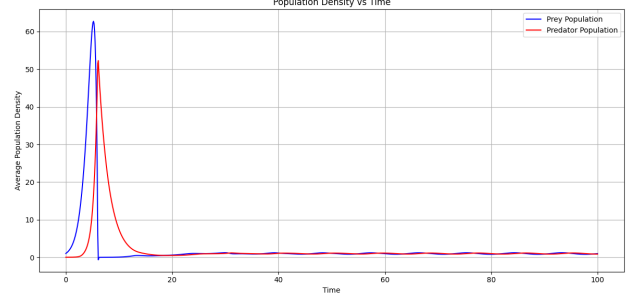


Figure 16: Populations density across time

In Figure 16, as prey was introduced first, it grows exponentially due to low predation. At $t = 5$, when the predator is introduced, the predator's graph shows a spike in population, due to high food sources. Then after a short delay the prey's population experiences a steep drop due to overpredation. The short lag was caused by the prey's reaction time to the increase in population growth of predator as it was a sudden introduction to the system. This lag represents the predator's biological time response in a stress-free environment where there are plenty of food and no external pressure.

Then, as the prey population is close to 0 the predator's population decreases showing its reliance on the prey's population. From the predator equation, when prey numbers, h , drop, the death term, $-\gamma l$, exceeds the birth term, $\delta h l$, the predator population declines. Overtime, as some surviving prey's reproduce the prey density increased slightly, which was followed by an increase in predator density. The system starts to demonstrate small oscillation that stabilises over time. Therefore, reaching dynamic equilibrium state.

The model can be validated with C.B. Huffaker's 1958 experiment [16] which showed that spatial heterogeneity and dispersal mechanism influence prey-predator interactions which ultimately lead to population oscillations and stabilisation.

3 Discussion and Conclusion

3.1 Discussion

This paper has analysed the predator-prey relationship between the Canadian lynx and Snowshoe hare, focusing on modifications to the assumptions that are associated with the basic Lotka-Volterra model [7, 10, 13]. The solutions gained from this report align with expectations: overall, each modified model improved the accuracy of predictions compared to historical data [6, 4].

Both Assumption 1 and Assumption 2 demonstrated improvements in model accuracy, as evidenced by the residuals found in the graphical representations of the systems. Similarly, Assumptions 3, 4, and 5 showed improvements through comparative analysis of the

graphs. These results indicate that refining the original assumptions enhances the model's ability to predict predator-prey interactions more accurately, and that having all these assumptions in the basic Lotka-Volterra model is detrimental to the overall accuracy. To further improve accuracy, we explored the possibility of combining different modified models. However, integrating all assumptions proved challenging. Assumption 1, in particular, was not easily combinable with other modifications. Assumption 4, on the other hand, showed improvements only when combined with Assumption 1, highlighting the interdependencies between certain modifications. Although, we were able to combine Assumption 3, 4 and 5.

3.2 Combining Assumptions

In this section, three assumptions were adopted: spatial (see Section 2.6), infinite food for prey (see Section 2.4) and infinite predator appetite (see Section 2.5). This model aims to closely fit to a real natural ecosystem, so that it enables the simulation of local prey-predator clustering, vegetation depletion, population waves, and real-life patterns absent in simpler models. These assumptions were represented by:

1. **Prey dynamics:** Growth and movement are influenced by vegetation availability.
2. **Predation:** Predators consume prey but with a limited appetite, modelled using Holling Type II functional response.
3. **Vegetation:** Grows independently at a constant rate but is consumed by prey, influencing prey population dynamics.

The equation below is **the modified prey and predator model**:

$$\frac{\partial h}{\partial t} = \alpha \left(\frac{u}{1+u} \right) h - \beta h l + D_h(t) \frac{\partial^2 h}{\partial x^2} \quad (28)$$

$$\frac{\partial l}{\partial t} = \delta \left(\frac{\beta h}{1+\beta h} \right) l - \gamma l + D_l(t) \frac{\partial^2 l}{\partial x^2} \quad (29)$$

Equation for the **time - dependent diffusion coefficients**:

$$D_h(t) = k \bar{u}(t) - k \left(\bar{h}(t) \frac{dc}{dt} - \frac{du}{dt} \right) t \quad (30)$$

$$D_h(t) \geq D_{\min} = 0.1 \quad (\text{minimum enforced}),$$

$$D_l(t) = 0.1 \cdot t$$

Equation for the **vegetation dynamics**:

$$\frac{\partial u}{\partial t} = \frac{du}{dt} - \frac{dc}{dt} h \quad (31)$$

3.2.1 Derivation of the Equation

[17] The mathematical model was constructed by analysing population interactions over space and time. Vegetation was incorporated into prey dynamics by

assuming prey reproduction depends on vegetation availability, replacing exponential growth, a saturation function limiting prey growth was used: $r \left(\frac{u}{1+u} \right) h - ahl$. However, in Assumption 5 (see Section 2.6), an alternative model was suggested. Both served the same purpose of simulating prey growth based on resource availability and limiting population growth to prevent unrealistic exponential increase. This model was adopted because it tracks vegetation $u(x, t)$ over time, allowing vegetation monitoring in the ecosystem. Assumption 4 (see Section 2.5) was applied, utilizing the finite appetite model for predator reproduction. $b \left(\frac{ah}{1+ah} \right) l$.

In the vegetation dynamics equation, $\frac{\partial u}{\partial t}$, the vegetation regeneration rate was assumed to be constant $\frac{du}{dt} = du_{dt}$, while prey consumption reduces it at a rate $-\frac{dc}{dt} h$.

Diffusion coefficients vary with time to reflect movement patterns. Prey diffusion $D_h(t)$ depends on vegetation distribution and prey density, with higher vegetation promoting greater dispersion. A minimum diffusion value $D_{\min}$ was proposed to prevent negative diffusion. Predator diffusion $D_l(t)$, unaffected by vegetation, increases linearly over time as $D_l(t) = 0.1 \cdot t$.

3.2.2 Results

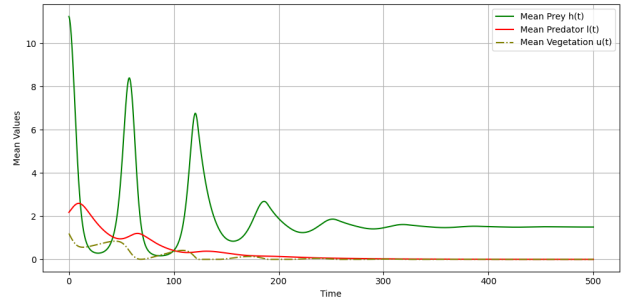


Figure 17: Graph of mean density values of prey, predator, vegetation varies along time.

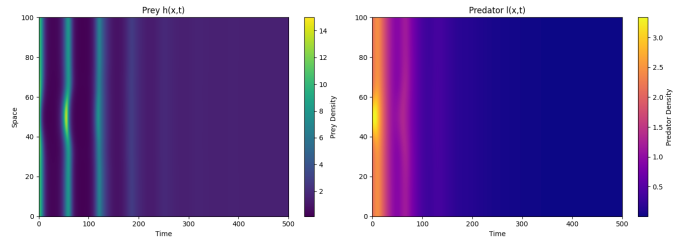


Figure 18: Heatmap of the Spatial diffusion of prey and predator combined with finite food for prey and predator having finite appetite.

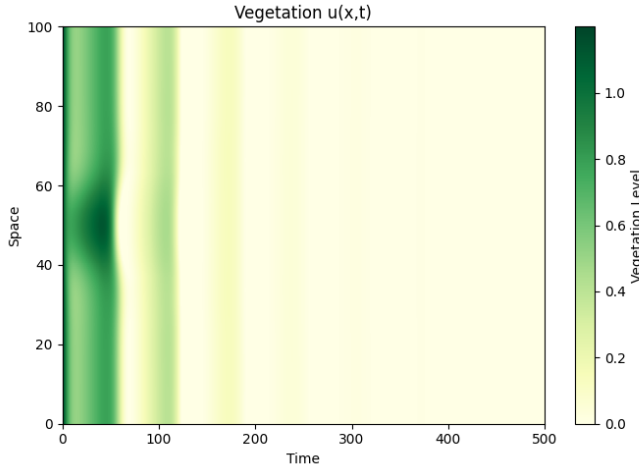


Figure 19: Heatmap of how vegetation changes across time and space.

In Figure 19, 17, 18, the graphs of prey, predator, and vegetation density show an ecosystem collapsing due to a lack of resources. At first, from $t = 0$ to $t = 70$, there is high vegetation throughout the space. As prey is introduced, it starts to consume the vegetation, leading to an increase in mean prey values. This results in growth in predator density with a time delay, as observed in the predator heat map and the mean curve. However, since the rate of prey consumption is greater than the vegetation growth rate, vegetation starts to deplete, causing lighter colours on the heatmap. This reduces both prey reproduction and diffusion rate $D_h(t)$, as the diffusion coefficient is influenced by vegetation and prey values. Vegetation can't regrow fast enough, causing a decline in both vegetation and prey, which then leads to a drop in predator population.

In the second phase, $t = 70$ to $t = 200$, the prey population experiences oscillations, representing attempts to recover areas with vegetation. On the prey heatmap, faint bands spread from the centre but weaken over time due to lack of vegetation. Vegetation remains low due to constant consumption and insufficient time for regeneration. This keeps prey density consistently low, while predators persist with minor peaks before steadily declining.

From $t = 200$ to $t = 500$, all three populations stabilise. Prey density flattens at a low value due to the lack of vegetation, limiting prey growth. Also, the diffusion coefficient $D_h(t)$ decreases over time, reaching the minimum value $D_{\min} = 0.1$, which restricts prey mobility and spatial redistribution. Predator values reach extinction, as the predator growth term becomes ineffective due to insufficient prey to sustain reproduction. With a constant death rate, predator density declines to zero. Vegetation is dominated by consumption, preventing meaningful regrowth.

Comparing between this model in Figure 17 with the spatial-only model in Figure 16, it becomes clear that incorporating a finite predator appetite results in more

moderate growth and reveals realistic patterns absent in the simpler model.

3.3 Limitations and Potential Improvements

This study reinforces that the assumptions inherent in the traditional Lotka-Volterra model are often oversimplifications of complex ecological interactions. While useful as a theoretical foundation, the model does not fully account for environmental and external factors that influence predator-prey dynamics. Real-world ecosystems are affected by variables such as climate fluctuations, habitat changes, and resource availability [18, 12, 11]. Certain assumptions were found to be particularly relevant in specific scenarios; for instance, assumption 3 (see Section 2.6) is crucial in modelling the effects of extreme weather conditions on hare populations [5].

The importance of environmental complexity in predator-prey interactions was demonstrated by Hufaker's 1958 experiment [16], which showed that habitat structure and spatial distribution can prevent rapid extinction and promote coexistence. This suggests that incorporating spatial heterogeneity and environmental factors into predator-prey models could further improve predictive accuracy.

Therefore, future research should explore environmental factors and additional external factors that may impact the predator-prey cycle. Collecting and incorporating our own data, particularly for assumption 3, would enhance model accuracy, though there are still issues, such as the difficulty in measuring vegetation changes.

In conclusion, this study demonstrates that modifications to the Lotka-Volterra model can significantly improve its predictive accuracy.

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